

Flywheel Mobile Upgrade Handoff — Follow-Up Review

Review date: July 28, 2026

Reference build: `main` at `6600cc3`

Previous review baseline: `fbd30ed8`

Purpose

This document translates the second-round mobile tester feedback into a prioritized implementation plan for the development team.

The recent updates have materially improved the game itself. The new flywheel avatar, stronger consumption feedback, clearer growth moments, denser city presentation, improved environmental rendering, corrected win logic, and new automated play bot all move the project in the right direction.

The primary remaining problem is no longer that the game lacks feel.

The primary remaining problem is that mobile players still encounter unnecessary friction before they can experience that improved gameplay.

The next development cycle should focus on:

1. Mobile touch-control clarity
2. First-session speed
3. Loading and input readiness
4. Genuine mobile test coverage
5. Mobile performance scaling
6. Accessibility and device-safe UI
7. Faster transitions between levels

Executive Summary

Current Assessment

Estimated mobile release readiness has improved from approximately **5/10** to **6.2/10**.

What improved

- The player character is more visually readable.
- Eating produces clearer feedback.

- Growth now feels rewarding.
- The city looks more deliberate and populated.
- Landmark objectives are easier to understand.
- Failure messaging now identifies the actual reason for failure.
- Desktop gameplay automation is significantly stronger.
- Level-completion confidence has improved.

What remains unresolved

- A first touch on the right half of the screen still controls the camera instead of movement.
- Fresh players still pass through several screens before Level 1.
- The title screen still contains excessive copy and instructions.
- The visible Start button may not be interactive until asynchronous asset loading completes.
- Existing browser tests still do not simulate real mobile touch behavior.
- No adaptive graphics profile has been implemented for weaker mobile devices.
- Mobile pause, mute, safe-area handling, and browser zoom remain incomplete.
- The upgrade shop continues to interrupt the next-level flow.

Preserve the Recent Improvements

The following changes were positively received and should be retained.

Player and Growth Feedback

Keep the new:

- Flywheel player model
- Growth shockwave
- Eating rim response
- Aggregated mass indicators
- Score-bar feedback
- Size-tier feedback
- Growth audio
- Reduced visual-noise approach

These improvements made growth feel like an event rather than a silent number change.

Objective Communication

Keep the unified win-condition system and its UI outputs:

- Landmark status chip
- Shield status
- Portal combo status
- “Grow bigger” status

- “Eat the landmark” ready state
- Progress-bar gating while the landmark remains
- Accurate failure explanations

The game should never again tell a player that they failed the mass goal when they actually failed a separate landmark condition.

City Presentation

Preserve the work completed around:

- Prop placement
- Building density
- Road clearance
- Prop proportions
- Ground-surface detail
- Building facades
- Lighting
- Shadows
- Atmosphere
- Horizon presentation

Future performance optimization should reduce quality gracefully rather than removing the game’s new visual identity across all devices.

Player-Fidelity Automation

Keep the closed-loop browser play bot.

It is substantially more valuable than a simple scripted movement test because it attempts to complete actual levels using normal input instead of directly changing game state.

The next step is to extend that philosophy to touch controls.

P0 — Required Before a Confident Mobile Release

1. Make First-Touch Movement Unambiguous

Current Behavior

The current mobile input scheme assigns roles according to where the first touch begins:

- Left half: virtual joystick and movement
- Right half: camera orbit

- Second finger: camera orbit

This is technically consistent, but it conflicts with natural one-handed mobile behavior.

A right-handed player commonly places their thumb in the lower-right portion of the screen. Under the current system, that first gesture rotates the camera instead of moving the player.

The player may interpret this as:

- Broken controls
- Input lag
- A frozen character
- A camera bug
- An unclear tutorial

Required Change

The first active touch should control movement regardless of whether it begins on the left or right side.

Recommended Control Scheme

- First finger: movement
- Second finger: camera orbit
- Two-finger gesture: camera pitch, where retained
- Optional visible camera-reset button
- Optional dedicated camera region only after the player has learned movement

Alternative

A fixed visible joystick may be used, but only when:

- It is clearly visible before interaction
- It remains reachable in both orientations
- It respects safe-area boundaries
- The camera region is also visibly communicated

Acceptance Criteria

- A first touch beginning in the lower-right quadrant moves the player.
- A first touch beginning in the lower-left quadrant moves the player.
- A single-finger drag cannot unexpectedly become camera orbit.
- Camera orbit requires a second finger or a visible dedicated control.
- Releasing movement input stops the player within 300 milliseconds.
- `pointerCancel`, browser blur, orientation changes, and interrupted gestures clear the input state safely.
- Movement continues correctly while a second finger adjusts the camera.
- Releasing the camera finger does not cancel the movement finger.
- The control scheme works in portrait and landscape.

- Unit tests cover all role transitions.
 - Browser tests use real touch events.
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2. Send Fresh Players Directly Into Level 1

Current Flow

A new player currently goes through:

1. Title screen
2. Start button
3. World map
4. First level selection
5. Level-introduction overlay
6. Dive In button
7. Gameplay

This delays the core experience and asks the player to understand progression systems before they understand the game.

Required Change

Fresh players with no completed levels should enter Level 1 directly after pressing Play.

Recommended First-Session Flow

1. Title screen
2. Play
3. Level 1 begins
4. Contextual movement instruction
5. First object consumed
6. Contextual growth instruction
7. Level completion
8. World map and upgrade systems introduced

Returning-Player Flow

Returning players may continue to enter through the world map or a Continue button.

Acceptance Criteria

- A fresh save reaches controllable gameplay within two taps.
- The first world-map visit occurs after completing or exiting Level 1.
- The first level does not require a separate level-selection action.
- The first level does not require an additional intro confirmation.
- Returning saves preserve normal world-map access.

- Existing save files remain compatible.
 - Automated tests verify fresh and returning flows separately.
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3. Fix the Visible-but-Unresponsive Start Button Risk

Current Behavior

The title screen and Start button are present in static HTML.

However, the Start button's click handler is attached only after the game waits for city textures and the model kit to load.

On a slower mobile connection or device, the player may see a button that appears ready but does nothing when tapped.

Required Change

Choose one of these approaches.

Preferred Approach

Wire the Start button immediately.

Begin noncritical asset loading in parallel and allow the game to fall back to procedural or lower-detail assets when needed.

Acceptable Alternative

Disable the button until initialization is complete and display an explicit loading state.

Example:

Preparing city...

Then enable and relabel the button as:

Play

Acceptance Criteria

- The Start button never appears enabled without a click handler.
- A tap during loading cannot be silently ignored.
- Initialization failure produces a usable fallback or visible error.
- Missing optional textures or models do not prevent gameplay.
- A slow-network browser test verifies button behavior.

- The loading state is screen-reader accessible.
 - The game records no duplicate Start actions from repeated taps.
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4. Replace Viewport-Only Mobile Tests With Real Touch Tests

Current Problem

The existing test suite contains mobile-sized viewports, but the primary gameplay flow remains desktop-oriented and keyboard-driven.

A narrow viewport is not equivalent to a mobile device.

Required Test Configuration

Use Playwright mobile emulation with:

- `isMobile: true`
- `hasTouch: true`
- A mobile user agent
- Realistic device scale factor
- Touchscreen events
- Portrait and landscape profiles

Required Profiles

At minimum:

- iPhone-sized Safari/WebKit-style profile
- Mid-range Android Chrome profile
- Small Android portrait profile
- Landscape phone profile

Required Test Scenarios

Fresh-Player Funnel

- Clear local storage.
- Open the game.
- Confirm the Start button is interactive.
- Tap Play.
- Confirm Level 1 begins directly.
- Confirm the HUD is visible.
- Confirm no map or intro overlay blocks gameplay.

First-Touch Movement

- Touch the lower-right quadrant.
- Drag upward.
- Confirm the avatar moves.
- Confirm the camera does not unexpectedly orbit.
- Release.
- Confirm movement stops.

Second-Finger Camera

- Begin movement with one finger.
- Add a second finger.
- Orbit the camera.
- Confirm movement continues.
- Release the second finger.
- Confirm the first finger still controls movement.

First Consumption

- Navigate toward an edible object.
- Confirm the object is consumed.
- Confirm mass increases.
- Confirm growth or eat feedback appears.
- Confirm the HUD updates.

Landmark Objective

- Reach or simulate the mass threshold while the landmark remains.
- Confirm the bar does not display a completed state.
- Confirm the landmark chip states the blocker.
- Confirm failure copy reports the correct reason.

Orientation

- Begin gameplay in portrait.
- Rotate to landscape.
- Confirm the renderer resizes.
- Confirm controls remain active.
- Confirm HUD elements remain reachable.
- Confirm safe-area padding is recalculated.

Browser Backgrounding

- Start a level.
- Background the page.
- Resume it.
- Confirm the timer did not unfairly continue.
- Confirm all touches were cleared.

- Confirm gameplay resumes safely.

Acceptance Criteria

- Touch E2E tests run on every pull request.
 - No keyboard input is used in mobile-control tests.
 - A right-side first touch is explicitly tested.
 - Portrait and landscape tests are both required.
 - Failures block merging.
 - Screenshots are stored for major mobile states.
 - Test results are available in CI.
 - The closed-loop play bot remains available as a separate deeper test.
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5. Add Adaptive Mobile Graphics Profiles

Current Risk

The updated visuals are better, but they increase the importance of performance validation.

Current rendering includes high-cost features such as:

- Antialiasing
- Dynamic directional shadows
- A 2048 × 2048 shadow map
- Detailed ground rendering
- Procedural facades
- Atmospheric effects
- A substantial triangle count
- Multiple gameplay feedback effects

This may work well on desktop and flagship phones while performing poorly on budget or thermally constrained Android devices.

Required Change

Implement quality tiers.

Recommended Profiles

High

- Device pixel ratio capped at 2
- Full 2048 shadow map
- Full shadow distance
- Antialiasing enabled
- Full decorative effects

- Full ground and facade detail

Medium

- Device pixel ratio capped at 1.5
- 1024 shadow map
- Reduced shadow distance
- Reduced particles and feedback density
- Reduced decorative animation
- Antialiasing based on device performance

Low

- Device pixel ratio capped at 1
- 512 shadow map, blob shadows, or no dynamic shadows
- Reduced particles
- Reduced animated effects
- Reduced facade or ground-detail resolution
- More aggressive distance culling
- Simplified rival and atmosphere effects where necessary

Runtime Performance Monitoring

Measure:

- Rolling average frame time
- p95 frame time
- Sustained FPS
- Long frames
- Current quality tier
- Device pixel ratio

Recommended downgrade behavior:

- Detect sustained poor performance.
- Reduce one quality tier.
- Do not repeatedly switch quality during active gameplay.
- Store the resulting tier locally.
- Allow manual override in Settings.

Acceptance Criteria

- A default quality tier is selected automatically.
- The quality tier can change without restarting the application.
- Low mode avoids the full 2048 shadow configuration.
- Quality choice persists.
- Development builds expose FPS and frame-time data.
- A manual quality setting is available.
- Performance is tested during large eating chains, rivals, landmark consumption, and city transitions.

- The game remains functional when optional visual assets fail.
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P1 — High-Value Mobile Improvements

6. Reduce the Opening Screen by Approximately 70%

Current Problem

The opening screen still explains:

- The premise
- The 3D format
- The number of cities
- The number of levels
- Movement
- Keyboard input
- Combos
- Goldens
- Rivals
- Landmarks
- Mute controls
- An easter egg

Most of this is irrelevant before the player has moved.

Required Change

Keep only:

- Game title
- One-line premise
- Play button
- Sound control
- Required attribution

Recommended Copy

FLYWHEEL

Swallow the city. Grow bigger. Devour the landmark.

PLAY

Acceptance Criteria

- The Play button is visible without scrolling at 360 × 640.
 - The title screen contains no more than two short descriptive sentences.
 - Keyboard instructions do not appear in primary mobile onboarding.
 - Advanced mechanics are introduced contextually.
 - The video does not delay input readiness.
 - Reduced-motion users receive a static title screen.
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7. Add Mobile Pause and Sound Controls

Required Controls

Add visible touch-accessible controls for:

- Pause
- Sound on or off

The pause menu should contain:

- Resume
- Restart level
- Sound
- Music, when separate
- Graphics quality
- Controls
- Return to map

Acceptance Criteria

- Touch targets are at least 44 × 44 CSS pixels.
 - Pausing stops gameplay and the timer.
 - Browser backgrounding automatically pauses or safely suspends the run.
 - Sound state persists.
 - Controls remain reachable around notches and home indicators.
 - The menu works in portrait and landscape.
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8. Add Safe-Area Support

Use:

- `env(safe-area-inset-top)`
- `env(safe-area-inset-right)`
- `env(safe-area-inset-bottom)`
- `env(safe-area-inset-left)`

Apply safe-area padding to:

- HUD
- Timer
- Objective chip
- Minimap
- Notifications
- Pause control
- Joystick
- Bottom overlay buttons
- Toasts
- Level-complete actions

Acceptance Criteria

- No HUD content overlaps a notch.
 - Bottom actions clear the iOS home indicator.
 - Landscape controls avoid left and right cutouts.
 - Devices without safe-area insets retain the current spacing.
 - Automated screenshots cover representative safe-area devices.
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9. Improve Mobile Text and Zoom Accessibility

Required Change

- Use at least 14 pixels for important mobile gameplay text.
- Reconsider `user-scalable=no`.
- Ensure critical information does not rely on color alone.
- Add accessible labels to controls.
- Retain reduced-motion support.
- Confirm screen-reader announcements do not become excessive.

Acceptance Criteria

- Objective, timer, mass, and failure text remain readable at 390 pixels wide.
 - Interactive controls have accessible names.
 - The interface remains usable when text scaling is increased.
 - Reduced-motion mode removes nonessential shake and looping movement.
 - Muted audio does not hide required gameplay information.
 - Too-large objects and landmark states include text or icon signals.
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10. Make the Upgrade Shop Optional

Current Behavior

The Level Complete action opens the upgrade shop before the next level.

This interrupts players who want to continue immediately.

Required Change

The Level Complete screen should offer:

- **Next Level** as the primary action
- **Upgrade** as a secondary action
- **World Map** as a tertiary action

Acceptance Criteria

- A player can move from Level Complete to the next playable level with one tap.
 - The shop can be skipped.
 - The world map is not mandatory between ordinary district levels.
 - Metro-completion moments may retain larger progression screens.
 - Upgrade availability remains clearly communicated without blocking play.
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P2 — Recommended Follow-Up Work

11. Add a Touch-Based Closed-Loop Play Bot

The current closed-loop browser bot uses desktop input.

Create a mobile equivalent that:

- Runs in a touch-enabled browser context
- Uses touch gestures only
- Chooses edible targets
- Moves using the same first-finger scheme as a player
- Adjusts the camera with a second finger
- Never writes game state
- Reports divergence from the simulation

Add two profiles:

Clean Touch Player

- Precise gestures

- Minimal delay
- Accurate directional changes

Sloppy Touch Player

- Delayed reactions
- Finger drift
- Overshooting
- Occasional incorrect direction
- Imperfect camera movement
- Short interruptions

The sloppy profile is particularly valuable because perfect bots do not expose the same usability problems as real players.

12. Add Mobile Development Telemetry

Expose in development builds:

- FPS
- Average frame time
- p95 frame time
- Quality tier
- Device pixel ratio
- Draw calls
- Triangle count
- Visible instances
- Active particles
- Player input state
- Active touch IDs
- Assigned touch roles
- Viewport size
- Orientation
- Safe-area values
- Renderer memory statistics

This can be exposed through a toggleable developer HUD or the existing debug object.

Recommended Implementation Sequence

Sprint 1 — Controls and First Session

- Change first-finger behavior to movement.
- Preserve second-finger camera controls.

- Add touch-state unit tests.
- Route fresh saves directly into Level 1.
- Simplify the opening screen.
- Wire Start before asynchronous loading.
- Add visible loading behavior.

Sprint 2 — Mobile Automation

- Add touch-enabled Playwright profiles.
- Add right-side first-touch tests.
- Add portrait and landscape flows.
- Add orientation-change tests.
- Add browser-background tests.
- Run tests in CI.
- Add mobile screenshots.

Sprint 3 — Performance

- Add graphics tiers.
- Add runtime frame monitoring.
- Add persistent quality settings.
- Test shadows and DPR on real devices.
- Define minimum supported hardware.

Sprint 4 — Accessibility and Flow

- Add pause and mute.
- Add safe-area support.
- Increase small HUD text.
- Review browser zoom.
- Make the shop optional.
- Add direct next-level continuation.

Sprint 5 — Real-Player Validation

Conduct at least 10 unmoderated or lightly moderated sessions across:

- iPhone Safari
- Android Chrome
- Small-screen phone
- Large-screen phone
- Landscape phone
- Mid-range Android
- Lower-performance Android

Do not explain the controls before the first attempt.

Playtest Metrics

Record:

- Time from page load to active Start button
 - Time from Play tap to controllable gameplay
 - Number of taps before Level 1
 - Time to first movement
 - Time to first object consumed
 - Percentage of players whose first gesture moves the camera accidentally
 - Level 1 completion rate
 - First-session abandonment point
 - Average and p95 frame time
 - Lowest sustained FPS
 - Number of UI mis-taps
 - Number of orientation-related failures
 - Percentage of players who understand the landmark condition
 - Percentage of players who continue to Level 2
 - Percentage of players who voluntarily enter the shop
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Definition of Done

The mobile upgrade is ready for broader testing when:

- A fresh player reaches Level 1 within two taps.
 - A first touch from either side of the screen moves the player.
 - A second finger controls the camera without canceling movement.
 - The Start button is never visibly active without being functional.
 - Level 1 is completable using touch-only input.
 - Mobile touch tests run in CI.
 - Portrait and landscape flows both pass.
 - The game supports adaptive quality settings.
 - Low mode avoids the most expensive rendering configuration.
 - Pause and mute are available without a keyboard.
 - HUD and controls respect device safe areas.
 - Important gameplay text is readable on a small phone.
 - The player can continue directly to the next level.
 - Real-device testing shows no repeated “the controls are broken” response.
 - Real-device testing shows acceptable sustained performance on the agreed minimum device.
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Priority Checklist

P0

- [] Make first touch control movement from either side
- [] Reserve camera orbit for second finger or visible control
- [] Add touch-role transition tests
- [] Send fresh players directly into Level 1
- [] Remove mandatory first-level map selection
- [] Remove mandatory first-level intro confirmation
- [] Wire Start before optional asset loading
- [] Add an explicit loading state
- [] Add real touch-driven mobile E2E
- [] Test lower-right first-touch movement
- [] Test portrait and landscape
- [] Add adaptive graphics tiers
- [] Add runtime frame monitoring
- [] Validate on real mid-range phones

P1

- [] Reduce title-screen copy
- [] Add mobile pause
- [] Add mobile sound control
- [] Add safe-area padding
- [] Increase important HUD text
- [] Review browser zoom restrictions
- [] Make the upgrade shop optional
- [] Add one-tap Next Level
- [] Add mobile performance telemetry

P2

- [] Add closed-loop touch play bot
- [] Add sloppy touch-player profile
- [] Add haptic feedback
- [] Add deeper accessibility checks
- [] Run external mobile playtests
- [] Record first-session funnel metrics

Final Direction

The recent work should be treated as a successful gameplay and visual-fidelity pass.

The team should not spend the next cycle rebuilding the avatar, adding more effects, expanding the meta systems, or increasing environmental complexity.

The next cycle should make it easier for a mobile player to reach, control, understand, and continue playing the game that now exists.

The most valuable result will not be another visual feature.

It will be a new player placing one thumb on the screen, moving immediately, swallowing their first object, understanding why they grew, and reaching Level 2 without confusion.